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RESEARCH ARTICLE

ON NUMERICAL SEMIGROUPS

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INTRODUCTION

This paper is concerned with the structure of numerical semigroups i.e. sets of linear combinations $n_1s_1+n_2s_2+\ldots+n_ts_t$ of a given set $\{s_i\}$ of natural numbers with non-negative integers n_i as coefficients.

Sylvester [17] stated and solved the following problem: "Let s_1 and s_2 be two relatively prime natural numbers. Determine the largest integer g which can not be written as a linear combination $n_1s_1+n_2s_2$ where n_1 and n_2 are non-negative integers". It is not hard to see that there exists such an integer g. The answer is $g=s_1s_2-s_1-s_2$. Moreover, as Sylvester showed, the interval $[0,g]$ contains equally many integers which can be written as non-negative linear combinations of s_1 and s_2 as integers which can not. We will refer to this last quality by saying that the semigroup generated by s_1 and s_2 is symmetric.

A natural generalization of Sylvester's problem was proposed by Frobenius (according to Brauer [1] "Frobenius mentioned it occasionally in his lectures"): "Let $s_1,s_2,\ldots,s_t$ be natural numbers with greatest common divisor 1. Determine (or bound) $g(s_1,s_2,\ldots,s_t)$, the largest integer which can not be written as $n_1s_1+n_2s_2+\ldots+n_ts_t$ for some non-negative integers $n_1,n_2,\ldots,n_t$." This number is now often referred to as the Frobenius number, and much work has been done to get an upper bound or (in special and rather rare cases) an exact expression for it. See e.g. [1], [2], [3], [4], [5], [6], [8], [9], [10], [11], [12],

[13], [15], and [16]. An overview of results in this area can be found in [14 , ch II].

In this paper we mainly study a concept which we call the type of the semigroup. The type of a semigroup is 1 if and only if the semigroup is symmetric. Our results make it possible to recover and generalize lots of results from the papers mentioned above. Among other things we classify the symmetric semigroups with three generators and we show that the type of a three-generated semigroup is at most two. We also solve a problem proposed by Selmer [13] to determine those semigroups which can be extended to a larger semigroup with the same Frobenius number. As a consequence we get that the number of different symmetric semigroups with given Frobenius number grows exponentially with the Frobenius number. We also consider a subclass of symmetric semigroups, called supersymmetric, which are particularly well behaved and classify them.

NUMERICAL SEMIGROUPS

Let $\mathbb{N}$ denote the natural numbers $\{0,1,2,\ldots\}$. A subsemigroup S of $(\mathbb{N},+)$ with $0 \in S$ is called a numerical semigroup. Each semigroup S has a natural partial ordering $\leq_S$ where for two elements s and t in S we have $s \leq_S t$ if there is a u in S so that $t=s+u$. The set $\{s_i\}$ of minimal elements in $S-\{0\}$ in this ordering is called a minimal set of generators for S. The reason is of course that all elements in S are linear combinations with non-negative integer coefficients of minimal elements. The set $\{s_i\}$ of minimal generators is finite since for any $s \in S$, $s \neq 0$, we must have $s_i \not\equiv s_j \pmod{s}$ if $i \neq j$. The same argument shows that the number of minimal generators is at most $\min\{s \in S;\ s \neq 0\}$. We denote the semigroup generated by $s_1,s_2,\ldots,s_t$ (i.e. the set $\{n_1s_1+n_2s_2+\ldots+n_ts_t;\ n_i \in \mathbb{N}$ for $i=1,2,\ldots,t\}$) by $\langle s_1,s_2,\ldots,s_t\rangle$. The semigroup $\langle s_1,s_2,\ldots,s_t\rangle$ is isomorphic to the semigroup $\langle ds_1,ds_2,\ldots,ds_t\rangle$ for any $d \in \mathbb{N} - \{0\}$. Thus when studying semigroups we can just as well assume that the greatest common

divisor of the generators is one. This is easily seen to be equivalent to that the semigroup has a finite complement in $\mathbb{N}$.

NOTE. In the sequel we use the word semigroup for numerical semigroup and we will furthermore always assume that semigroups have finite complement in $\mathbb{N}$ unless the opposite is explicitely stated.

DEFINITION. For a semigroup S we denote $\max\{x \in \mathbb{Z};\ x \notin S\}$ by $g(S)$. This number is often called the Frobenius number of S.

For a semigroup S we denote by $g-S$ the set of numbers $\{g(S)-s;\ s \in S\}$. The two sets S and $g-S$ are disjoint since $g - s \in S$ would imply $(g-s) + s = g \in S$.

DEFINITION. The semigroup S is called symmetric if $S \cup (g-s) = \mathbb{Z}$. There are several alternative descriptions of the concept of symmetric semigroups.

LEMMA 1. The following conditions are equivalent for a semigroup S.

(a) S is symmetric.

(b) For each $x \in \mathbb{Z}$ we have $x \in S$ or $g(S) - x \in S$.

(c) There is a number $a \notin S$ such that for each $x \in \mathbb{Z}$ we have $x \in S$ or $a - x \in S$.

(d) If $x+y = g(S)$ then exactly one of x and y belongs to S.

(e) There is a number $a \notin S$ such that $x+y = a$ implies that exactly one of x and y belongs to S.

(f) Among the numbers $0,1,\ldots,g(S)$ there are just as many elements in S as there are elements outside S.

PROOF. Remembering that S and $g-S$ are always disjoint we see that (b) and (d) are just reformulations of (a). Of course (b) implies (c) and since no negative number x belongs to S we must have that a is the largest number not belonging to S hence (c) implies (b),

and (e) is just a reformulation of (c). Since S symmetric is equivalent to that for each $x \in \mathbb{Z}$ exactly one of x and $g-x$ belongs to S and since this condition is always true for $x < 0$ we get that (f) is equivalent to symmetry. Note that this shows in particular that $g(S)$ must be an odd number if S is symmetric.

Next we shall for each semigroup S define a set S' which will be of great importance in what follows.

DEFINITIONS. For a semigroup S we set $S' = \{x \in \mathbb{Z};\ x \notin S,\ x + s \in S$ for all $s \in S,\ s > 0\}$. The number of elements in S' will be called the type of S.

We see that for a semigroup S we always have $g(S) \in S'$. This is the only number which belongs both to S' and $g-S$ since if $x = g-s$, $s > 0$, then $x + s \notin S$. This shows that a symmetric semigroup must have type equal to one.

DEFINITION. For a semigroup S which is not symmetric we set $h(S) = \max\{x;\ x \notin S,\ x \in g-S\}$.

It is not hard to see that if $h(S)$ is defined, i.e. if S is not symmetric, then $h(S) \in S'$. For suppose that for some $s > 0$, $s \in S$ we have $h(S) + s \notin S$. The maximality of $h(S)$ would then imply that $h(S)+s \in g-S$, say $h(S) + s = g(S)-s'$, $s' \in S$. But then $h(S) = g(S) - (s'+s) \in g - S$ which yields a contradiction. This shows in particular that if S is not symmetric then the type is at least two. We have proved the following proposition.

PROPOSITION 2. The semigroup S is symmetric if and only if type$(S) = 1$, i.e. if and only if $S' = \{g(S)\}$.

This characterization of symmetric semigroups will be the most useful to us in the sequel. Next we shall consider semigroups which are not symmetric and in a similar way characterize a class of non-symmetric semigroups which

behaves as the symmetric ones in some respects.

It follows from above that if S is not symmetric then $h(S)$ is the second largest element in S'. It also follows that $2h(S) \geq g(S)$ since if $x \notin S$, $x \notin g - S$, then $g(S) - x \notin S$ and $g(S) - x \notin g - S$ and for each $x \notin Z$ at least one of x and $g(S) - x$ is greater than on equal to $g(s)/2$.

LEMMA 3. Suppose that S is a semigroup with $g(S)$ an even number. Then the following conditions are equivalent.

(a) $S \cup (g-S) = \mathbb{Z} - \{g(S)/2\}$.

(b) For each $x \in \mathbb{Z}$ we have $x \in S$ or $x \in g - S$ or $x = g(S)/2$.

(c) $S' = \{g(S)/2, g(S)\}$.

(d) Among the numbers $0,1,\ldots,g(S) - 1$ there are just as many elements in S as there are elements outside S.

PROOF. The conditions (a) and (b) are just reformulations of each other and that these are equivalent to (d) follows as in Lemma 1. Assume (a). Since S' has only g in common with $g - S$ and has no element in common with S we get that $S' \subseteq \{g(S)/2, g(S)\}$. But S is not symmetric since $g(S)$ is even hence $S' = \{g(S)/2, g(S)\}$ and (a) implies (c). We have seen that $h(S)$ is the second largest element in S' and is the largest element outside both S and $g - S$ by definition. Thus (c) implies (a).

Next we want to study what happens with the Frobenius number when a semigroup is enlarged.

DEFINITION. Let $S = \langle s_1, s_2, \ldots, s_t \rangle$. For $a \in \mathbb{N}$ we denote $\langle s_1, s_2, \ldots, s_t, a \rangle$ by $\langle S, a \rangle$.

Selmer [13, section 4] proposes the problem to decide for which semigroups S there exists an $a \notin S$ such that $g(S) = g(\langle S, a \rangle)$. Mendelsohn [8] noticed that no such a exists if S is generated by two elements and the case of three-generated semigroups is treated in [7]. The

following proposition gives an answer to the problem.

PROPOSITION 4. Let g be a fixed number. Consider $\mathcal{S}_g = \{S;\ S$ a semigroup with $g(S) = g\}$. The elements in $\mathcal{S}_g$ are partially ordered with respect to inclusion.

(a) If g is odd we have that $S \in \mathcal{S}_g$ is maximal in $\mathcal{S}_g$ if and only if S is symmetric.

(b) If g is even we have that $S \in \mathcal{S}_g$ is maximal in $\mathcal{S}_g$ if and only if $S' = \{g/2, g\}$. (In particular type$(S) = 2$.)

PROOF. (a) If $S \in \mathcal{S}_g$ is symmetric and $a \notin S$ we have $a = g - s$ for some $s \in S$. Thus $g = a + s \in \langle S, a \rangle$ so $g(\langle S, a \rangle) < g$ and hence S is maximal in $\mathcal{S}_g$. If S is not symmetric we have $g \notin \langle S, h(S) \rangle$ since if $g = s + nh(S)$ for some $s \in S$ and some $n \in \mathbb{N}$ we must have $n=1$ since $2h(S) > g$ and thus this would give $h(S) = g - s$, a contradiction. Thus S is not maximal in $\mathcal{S}_g$.

(b) Suppose $S' = \{g/2, g\}$. If $a \notin S$ we have $a \in g - S$ or $a = g/2$. In both cases we get $g \in \langle S, a \rangle$, thus $g(\langle S, a \rangle) < g$ and S is maximal in $\mathcal{S}_g$. Suppose $S' \neq \{g/2, g\}$. Then $h(S) > g/2$ and we can conclude as in (a) that $g(\langle S, h(S) \rangle) = g$ and that S is not maximal in $\mathcal{S}_g$.

EXAMPLE. Proposition 4 shows that each semigroup S can be embedded in a semigroup S_1 with $g(S) = g(S_1)$ and such that S_1 is symmetric (if $g(S)$ is odd) or such that $S_1' = \{g(S)/2, g(S)\}$ (if $g(S)$ is even). Such an embedding is not unique in general. A simple example of the non-uniqueness is the following. If $S = \langle 4,6,7,9 \rangle$ we have $g(S) = 5$ and S can be embedded in both $S_1 = \langle 2,7 \rangle$ and $S_2 = \langle 3,4 \rangle$ which are symmetric with $g(S_1) = g(S_2) = 5$.

The next proposition shows that the number of symmetric semigroups S with $g(S) = g$ grows exponent-

ially with g .

PROPOSITION 5. Let g be a fixed odd number. The number of symmetric semigroups S with $g(S) = g$ is at least $2^{[g/8]}$, where $[g/8]$ denotes the integer part of $g/8$.

PROOF. We will show that the semigroup $T = \langle g+1, g+2, \ldots, 2g+1 \rangle$ (which has $g(T) = g$) can be embedded in at least $2^{[g/8]}$ different semigroups S with $g(S) = g$. First we can extend T to a semigroup T_1 by adjoining as generators any set E of even numbers between $[g/4] + 1$ and $\lceil g/2 \rceil$. Next, if T_1 is not symmetric, we extend T_1 to T_2 by adjoining the generator $h(T_1)$ and so on until we eventually reach a symmetric semigroup T . It follows easily that different choices of the set E yields different semigroups T and that the number $g(T_i)$ remains the same throughout the process. Since E can be chosen in at least $2^{[g/8]}$ ways this completes the proof.

The following device has been used by Brauer and Shockley [3]. We will use it to get an alternative description of the set S' belonging to a semigroup S .

DEFINITION. For a semigroup S and any $s \in S$, $s \neq 0$ we let $S(s) = \{t \in S;\ t - s \notin S\}$.

Clearly we have that $t \in S(s)$ if and only if t is the smallest number in S in some residue class modulo s. Thus, as S has a finite complement in $\mathbb{N}$, there are s elements in $S(s)$.

LEMMA 6. The set $S(s)$ is an isolated subset of S in the partial ordering $\leq_S$, i.e. if $t \in S(s)$ and $u \leq_S t$ in S then $u \in S(s)$.

PROOF. This is immediate from the definitions.

PROPOSITION 7. Let S be a semigroup and $s \in S$, $s \neq 0$. Then the following conditions on an integer t are

equivalent.

1) $t - s \in S'$

2) t is maximal in $S(s)$ with respect to the partial ordering $\leq_S$.

PROOF. $t - s \in S' \Leftrightarrow t - s \notin S$ but $t - s + u \in S$ for all $u \in S$, $u > 0 \Leftrightarrow t \in S(s)$ but $t + u \notin S(s)$ for all $u \in S$, $u > 0 \Leftrightarrow t$ is maximal in $S(s)$ with respect to the partial ordering $\leq_S$.

COROLLARY 1. If S is a semigroup and $s \in S$, $s \neq 0$, then the type of S equals the number of maximal elements in $S(s)$ with respect to the partial ordering $\leq_S$.

COROLLARY 2. The type of a semigroup S is less than or equal to $\min\{s \in S;\ s \neq 0\} - 1$.

PROOF. Let s be the least positive number in S . Since $0 \in S(s)$ we have that $S(s)$ contains at most $s-1$ maximal elements.

COROLLARY 3. (Brauer - Shockley, [3]). If S is a semigroup and $s \in S$, $s \neq 0$ we have $g(S) = \max S(s) - s$.

PROOF. We have that $g(S)$ is the largest element in S'.

EXAMPLES. For the semigroups $\langle 2, 2k+1 \rangle$ and $\langle k, k+1, \ldots, 2k-1 \rangle$ there is equality in Corollary 2.

As another rather immediate consequence we get $g = s_1 s_2 - s_1 - s_2$ for the semigroup minimally generated by the two numbers s_1 and s_2 . This classical formula of Sylvester [17] was the starting point for research in Frobenius numbers of semigroups.

The same technique can be used to calculate $S(a)$ (and thereby S' and $g(S)$) for semigroups S generated by an arithmetic sequence of numbers $a, a+d, a+2d, \ldots, a+kd$. Thus we can rediscover and generalize results in [10] (also c.f. [2], [4] and [13]). In this case the type of S equals the remainder between 1 and k , when $a-1$ is

divided by k .

PROPOSITION 8. Let d and s_t be relatively prime natural numbers. Let $S = \langle s_1, s_2, \ldots, s_t \rangle$ and $T = \langle ds_1, ds_2, \ldots, ds_{t-1}, s_t \rangle$. Then we have

(a) $T(s_t) = \{ds;\ s \in S(s_t)\}$.

(b) The type of S equals the type of T . In particular we have that S is symmetric if and only if T is symmetric.

(c) (Johnson, Brauer and Shockley) $g(T) = dg(S) + (d-1)s_t$.

PROOF. Let $t \in T(s_t)$. Then $t \in \langle ds_1, ds_2, \ldots, ds_{t-1} \rangle$, say $t = ds$, where $s \in S$. We have $t - s_t \notin T$ and hence $ds - ds_t \notin T$. Thus $s - s_t \notin S$ i.e. $s \in S(s_t)$. Thus every element in $T(s_t)$ is of the form ds , where $s \in S(s_t)$. The equality in (a) follows since $T(s_t)$ and $S(s_t)$ both have s_t elements. Then (b) and (c) follow from Corollary 1 and Corollary 3 to Proposition 7.

DEFINITION. Let $S = \langle s_1, s_2, \ldots, s_t \rangle$ and let for each i the largest common divisor of $\{s_1, s_2, \ldots, \hat{s_i}, \ldots, s_t\}$ be denoted by d_i . We call the semigroup $\bar{S}$ generated by $\{s_i / \prod_{j \neq i} d_j;\ i = 1,2,\ldots,t\}$ the derived semigroup of S .

COROLLARY. Let S be a semigroup and $\bar{S}$ its derived semigroup. Then S and $\bar{S}$ have the same type. In particular S is symmetric if and only if $\bar{S}$ is symmetric.

EXAMPLES. If $S = \langle s_1, s_2 \rangle$ we have $\bar{S} = \langle 1 \rangle = \mathbb{N}$ and once more we see that S is symmetric (since $\mathbb{N}$ is symmetric) and that $g\langle s_1, s_2 \rangle = s_2 g(\mathbb{N}) + (s_2-1)s_1 = -s_2 + (s_2-1)s_1$. If $T = \langle s_1, s_2, s_3 \rangle$ we see that when determining the type of T and $g(T)$ we can assume s_1, s_2, s_3 to be pairwise relatively prime.

PROPOSITION 9. Let $S = \langle s_1,s_2,\ldots,s_t \rangle$ be a semigroup. Then the following two conditions on an element $u \in \langle s_1,s_2,\ldots,s_{t-1} \rangle$ are equivalent.

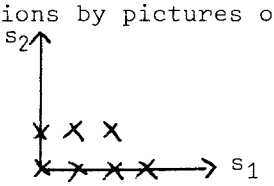
(1) $u \notin S(s_t)$ but $u - s_i \in S(s_t)$ if $s_i \leq_S u$ and i is between 1 and t-1 .

(2) u is the least positive multiple of s_t in $\langle s_1,s_2,\ldots,s_{t-1} \rangle$.

PROOF. The condition $u \notin S(s_t)$ is equivalent to $s_t \leq_S u$. Thus if u satisfies condition (1) we have $s_t \leq_S u$ but $s_t + s_i \not\leq_S u$ for $i = 1,2,\ldots,t-1$. It follows at once that $u = ks_t$ for some k . If we had $ms_t \in \langle s_1,s_2,\ldots,s_{t-1} \rangle$ for some $m < k$, $m \neq 0$ then for some i between 1 and t-1 we would have $s_i \leq_S ms_t$ and hence $s_i + s_t \leq_S u$. Thus k is minimal, i.e. u satisfies condition (2). Now suppose that $u = p \cdot s_t$ is the least positive multiple of s_t in $\langle s_1,s_2,\ldots,s_{t-1} \rangle$. Clearly, $u \notin S(s_t)$. Take any i between 1 and t-1 such that $s_i \leq_S u$ and choose r maximal such that $rs_t + s_i \leq_S u$. Then $0 < (p-r)s_t \in \langle s_1,s_2,\ldots,s_{t-1} \rangle$ and hence $r=0$. Thus $s_t + s_i \leq_S u$, and hence u satisfies condition (1).

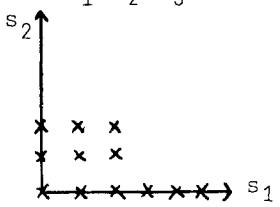
DEFINITIONS. The unique number u which satisfies the conditions of Proposition 9 is called the corner of $S(s_t)$. We denote by $L(s_t)$ the set of vectors $(\alpha_1,\alpha_2,\ldots,\alpha_{t-1})$ in $\mathbb{N}^{t-1}$ such that $\alpha_1 s_1 + \alpha_2 s_2 + \ldots + \alpha_{t-1} s_{t-1} \in S(s_t)$. An element $(\alpha_1,\alpha_2,\ldots,\alpha_{t-1})$ in $L(s_t)$ is called maximal in $L(s_t)$ if $(\alpha_1,\alpha_2,\ldots,\alpha_{i-1},\alpha_i+1, \alpha_{i+1},\ldots,\alpha_{t-1}) \in L(s_t)$ for $i = 1,2,\ldots,t-1$.

We exemplify the definitions by pictures of $L(s_t)$ for two cases. First let $\langle s_1,s_2,s_3 \rangle = \langle 5,6,7 \rangle$



Here $L(7) = \{(0,0),(1,0),(2,0),(3,0),(0,1),(1,1),(2,1)\}$ and $(3,0)$ and $(2,1)$ are maximal elements in $L(7)$. The corner of $S(7)$ is $3s_1+s_2 = 21 = 3\cdot 7$.

Next let $(s_1,s_2,s_3) = \langle 4,6,9 \rangle$.



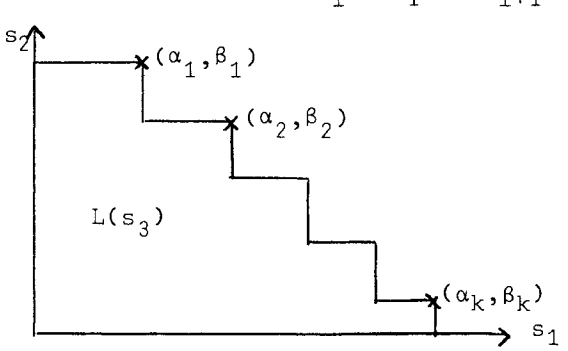
Here $L(9) = \{(0,0),(1,0),(2,0),(3,0),(4,0),(5,0),(0,1),(1,1),(2,1),(0,2),(1,2),(2,2)\}$.

The maximal elements are $(5,0)$ and $(2,2)$ and they represent the same element 20 in S. The corner of $S(9)$ is $3s_1+s_2 = 3s_2 = 18 = 2\cdot 9$.

We shall now for a while specialize to semigroups generated by three elements.

LEMMA 10. Let $S = \langle s_1,s_2,s_3 \rangle$. If there are at least three maximal elements in $L(s_3)$ then S is symmetric.

PROOF. Let the maximal elements in $L(s_3)$ be $(\alpha_1,\beta_1),(\alpha_2,\beta_2),\ldots,(\alpha_k,\beta_k)$ and assume $k \geq 3$. We can not have maximal elements in $L(s_3)$ with the same first coordinate, thus we can assume $\alpha_1<\alpha_2<\ldots<\alpha_k$. Then it follows that $\beta_1>\beta_2>\ldots>\beta_k$. Next we observe that for each $i = 1,2,\ldots,k-1$ we have that $(\alpha_i+1)s_1 + (\beta_{i+1}+1)s_2$ is the corner of $S(s_3)$.



Proposition 9(b) implies that $(\alpha_i+1, \beta_{i+1}+1)$ all represent the same element in $S(s_3)$ for $i=1,2,\ldots,k-1$ since the corner is unique. This implies of course that also $\alpha_i s_1+\beta_{i+1}s_2$ are all equal for $i = 1,2,\ldots,k-1$ and this gives that $(\alpha_i,\beta_{i+1}) + (0,\beta_1-\beta_2)$ all represent the same element for $i = 1,2,\ldots,k-1$. But for $i=1$ we get (α_1,β_1) which is a maximal element in $L(s_3)$, that is represent a maximal element in $S(s_3)$. Thus $(\alpha_i,\beta_{i+1}+\beta_1-\beta_2)$ represent the same maximal element in $S(s_3)$ hence $(\alpha_i,\beta_{i+1}+\beta_1-\beta_2)$ is a maximal element in $L(s_3)$ for $i = 1,2,\ldots,k-1$. This gives that $\beta_{i+1}+\beta_1-\beta_2 = \beta_i$ which shows that (α_i,β_i) all represent the same element in $S(s_3)$ for $i = 1,2,\ldots,k-1$. Finally, since $(\alpha_{k-2},\beta_{k-1})$ and (α_{k-1},β_k) represent the same element in $S(s_3)$ we get that $(\alpha_{k-2},\beta_{k-1}) + (\alpha_k-\alpha_{k-1},0)$ and $(\alpha_{k-1},\beta_k) + (\alpha_k-\alpha_{k-1},0) = (\alpha_k,\beta_k)$ represent the same element in $S(s_3)$ and the same reasoning as above shows that also (α_k,β_k) represent the same element as (α_i,β_i) for $i < k$. We have shown that all maximal elements in $S(s_3)$ are the same, hence the type of S is one, i.e. S is symmetric.

We will give two proofs of the following result, one using the concepts of corner of $S(s_3)$ and maximal elements in $L(s_3)$ we have just developed, the other more direct and avoiding those concepts.

THEOREM 11. If $S = \langle s_1,s_2,s_3 \rangle$ then the type of S does not exceed two.

FIRST PROOF. If the number of maximal elements in $L(s_3)$ is at least three, S is symmetric by Lemma 10. The type equals the number of maximal elements in $S(s_3)$ which is less than or equal to the number of maximal elements in $L(s_3)$.

SECOND PROOF. We first describe a way to construct elements in S' . Let $a = \max\{j;\ -s_3+js_1 \notin S\}$ and let $b = \max\{k;\ -s_3+as_1+ks_2 \notin S\}$. Then $z = -s_3+as_1+bs_2 \in S'$

since $z \notin S$ but $z+s_i \in S$ for $0 = 1,2,3$. We say that the element z constructed in this way is (s_3,s_1,s_2)-canonically generated. In this way we have a priori constructed six elements in S' by permuting (s_1,s_2,s_3). Now let $x \in S'$. Then x can be written as $x = a_2s_1-s_2+c_2s_3 = a_3s_1+b_3s_2-s_3$ for some non-negative integers a_2,c_2,a_3,b_3. We claim that x is (s_2,s_3,s_1)- or (s_3,s_2,s_1)-canonically generated, that is that $-s_2+(c_2+1)s_3 \in S$ or $-s_3+(b_3+1)s_2 \in S$. Suppose that $-s_2+(c_2+1)s_3 \notin S$ and $-s_3+(b_3+1)s_2 \notin S$. Then $x = a_2s_1-s_2+c_2s_3$ gives $-s_2+(c_2+1)s_3 = x+s_3-a_2s_1 = (a_3-a_2)s_1+b_3s_2$. Since this element does not belong to S we must have $a_3 < a_2$. Similarly $-s_3+(b_3+1)s_2 = x+s_2-a_3s_1 = (a_2-a_3)s_1+c_2s_3$ implies $a_2 < a_3$ and gives a contradiction. This proves that S' contains at most two elements.

NOTE. The following observation is due to J. Backelin: There is no upper bound on type $\langle s_1,s_2,\ldots,s_t \rangle$ for any $t \geq 4$.

EXAMPLE. Let $n \geq 2$ and $r \geq 3n+2$ be given and let $s = r(3n+2) + 3$ and $S = \langle s, s+3, s+3n+1, s+3n+2 \rangle$. Let $M = \{(r+1)s+i;\ -2 \leq i \leq 3n-1,\ i \not\equiv 0 \pmod 3\}$. We claim that $M \subseteq S'$. First we observe that if $x \in S$, $x < (r+2)s$, then x must be a sum of exactly $[x/s]$ generators. Then it is easy to check that $M \cap S = \emptyset$. After that one shows that $M+x \in S$ for each generator x of S. Thus $M \subseteq S'$ and type $S \geq 2n+2$. (In fact, a more careful analysis shows that $S' = M \cup \{(2n-1)s+6n\}$, hence type $S = 2n+3$.)

PROPOSITION 12. If $S = \langle s_1,s_2,s_3 \rangle$ is not symmetric then the elements in $S(s_3)$ have unique representations in $L(s_3)$.

PROOF. Suppose some element in $S(s_3)$ has two different representations in $L(s_3)$, say (α_1,β_1) and (α_2,β_2) . There are $\gamma,\delta \geq 0$ such that $(\alpha_1+\gamma,\beta_1+\delta)$ is a maximal element in $L(s_3)$, hence $(\alpha_2+\gamma,\beta_2+\delta)$ is an element in $L(s_3)$ since maximal elements in $L(s_3)$ correspond to maximal elements in $S(s_3)$ and $(\alpha_1+\gamma,\beta_1+\delta)$ must represent the same element in $S(s_3)$ as $(\alpha_2+\gamma,\beta_2+\delta)$ does. Thus there is a maximal element in $S(s_3)$ that corresponds to two different (maximal) elements in $L(s_3)$. If the number of maximal elements in $L(s_3)$ is at least three we have by Lemma 10 that S is symmetric. If the number of maximal elements in $L(s_3)$ is two we have shown that they represent the same element in $S(s_3)$, thus S is symmetric also in this case as of course if there is only one maximal element in $L(s_3)$.

NOTE. For symmetric semigroups $\langle s_1,s_2,s_3 \rangle$ we can get as many maximal elements in $L(s_3)$ as we want as the example $\langle s_1,s_2,s_3) \rangle = (4,6,2k+1)$ shows.

LEMMA 13. Let $S = \langle s_1,s_2,s_3 \rangle$ and assume that s_1 and s_2 are relatively prime and that $s_3 < s_1$ or $s_3 < s_2$. Then every element in $S(s_3)$ has unique coordinates in $L(s_3)$.

PROOF. Say $s_3 < s_1$ and both (n_1,n_2) and (m_1,m_2) represent the same element $n_1s_1+n_2s_2 = m_1s_1+m_2s_2$ in $S(s_3)$. Suppose $n_1 \geq m_1$ and hence $n_2 \leq m_2$. Then $(n_1-m_1)s_1 = (m_2-n_2)s_2 \leq_S n_1s_1+n_2s_2$ and hence belongs to $S(s_3)$ by Lemma 6. Thus it suffices to show that $as_1 = bs_2 \in S(s_3)$ implies $a=b=0$. Now if $bs_2 \in S(s_3)$ then $ks_2 \in S(s_3)$ for $k = 0,1,\ldots,b$ and hence $b < s_3 < s_1$. Since s_1 and s_2 are relatively prime we have that s_1 divides b , thus $b=0$ and hence $a=0$.

THEOREM 14. A semigroup minimally generated by three pairwise relatively prime elements is not symmetric.

PROOF. Let $S = \langle s_1,s_2,s_3 \rangle$ where $s_1 > s_3$ and $s_2 > s_3$ and s_1,s_2,s_3 pairwise relatively prime and suppose that S is symmetric. We aim to show that one of s_1 or s_2 is superfluos as generator, i.e. that $s_1 \in \langle s_2,s_3 \rangle$ or $s_2 \in \langle s_1,s_3 \rangle$. Our first step is to show that the corner of $S(s_3)$ is a multiple of s_1 or s_2. Suppose that $(a_1+1)s_1 + (a_2+1)s_2$ is the corner of $S(s_3)$. Then $S(s_3)$ would contain a maximal element $a_1s_1+b_2s_2$ for some $b_2 > a_2$ and a maximal element $b_1s_1+a_2s_2$ for some $b_1 > a_1$. By Lemma 13 these two maximal elements in $S(s_3)$ are different which contradicts the fact that S is symmetric. Thus the corner of $S(s_3)$ is a multiple of s_1 or s_2. Say the corner is $ks_3 = as_1$. Since s_1 and s_3 are relatively prime we get that s_3 divides a. From the minimality of k we get $a=s_3$ and thus $0,s_1,2s_1,\ldots,$ $\ldots,(s_3-1)s_1 \in S(s_3)$. But S is generated by $S(s_3)$ and s_3, thus $S = \langle s_1,s_3 \rangle$.

NOTE. If $\langle s_1,s_2,s_3 \rangle$ is symmetric where $s_1 < s_2 < s_3$ and s_1,s_2,s_3 pairwise relatively prime we have of course $s_3 \in \langle s_1,s_2 \rangle$ since obviously $s_1 \notin \langle s_2,s_3 \rangle$ and $s_2 \notin \langle s_1,s_3 \rangle$.

The following corollary gives an easy criterion for deciding if a three-generated semigroup is symmetric.

COROLLARY. The semigroup $S = \langle s_1,s_2,s_3 \rangle$ is symmetric if and only if its derived semigroup $\bar{S}$ is generated by two elements.

PROOF. The derived semigroup $\bar{S}$ is generated by three pairwise relatively prime elements.

To decide whether a semigroup $\langle s_1,s_2,s_3 \rangle$ is symmetric or not we suggest the following algorithm. First determine the derived semigroup $\langle \bar{s}_1,\bar{s}_2,\bar{s}_3 \rangle$ and

suppose $\bar{s}_3$ is the largest of these three elements. If $\bar{s}_3 > \bar{s}_1\bar{s}_2-\bar{s}_1-\bar{s}_2$ then S is symmetric. Otherwise let $c = [\bar{s}_3/\bar{s}_2]$. For $i = 1,2,\ldots,c$ determine if $\bar{s}_1$ divides $\bar{s}_3-i\bar{s}_2$. If this is so for some i then S is symmetric otherwise S is not symmetric.

We now turn back to semigroups with an arbitrary number of generators. We shall next single out a class of semigroups which turn out to be symmetric with extremely good behaviour in some respect. We start with an abstract definition but will classify all these semigroups quite concretely. We will however first need two lemmas and a preliminary definition.

DEFINITION. The semigroup $S = \langle s_1,s_2,\ldots,s_t \rangle$ is called s_t-supersymmetric if $-s_t+n_is_i \notin S$, $n_i \in \mathbb{N}$, $i = 1,2,\ldots,t-1$ implies that $-s_t+\sum_{i=1}^{t-1} n_is_i \notin S$.

LEMMA 15. The semigroup $S = \langle s_1,s_2,\ldots,s_t \rangle$ is s_t-supersymmetric if and only if it is symmetric and every element in $S(s_t)$ has a unique representation in $L(s_t)$.

PROOF. Suppose S to be s_t-supersymmetric. For each $i = 1,2,\ldots,t-1$ we set $a_i = \max\{n_i;\ -s_t+n_is_i \notin S\}$. Then $-s_t + \sum_{i=1}^{t-1} a_is_i \notin S$ and must be the unique maximal element $g(S)$ in S' . Thus S is symmetric. We have seen that if there is an element in $S(s_t)$ with two different representations in $L(s_t)$, then there is a maximal element in $L(s_t)$ with two different representations, c.f. the proof of Proposition 12 which on this point is valid for semigroups with more than 3 generators. But $L(s_t)$ obviously has the unique maximal element $(a_1,a_2,\ldots,a_{t-1})$. Now suppose that S is symmetric and that elements in $S(s_t)$ have unique representations in $L(s_t)$. Let $(a_1,a_2,\ldots,a_{t-1})$ be the unique maximal element in $L(s_t)$ hence $g(S) = -s_t + \sum_{i=1}^{t-1} a_is_i$. Then

of course $-s_t+a_is_i \notin S$ for $i = 1,2,\ldots,t-1$. Suppose that $-s_t+b_js_j \notin S$ for some j and that $a_j < b_j$. Then there would be a maximal element $-s_t + \sum_{i=1}^{t-1} c_is_i$ in $S(s_t)$ with $c_j \geq b_j$, which would contradict the uniqueness of the representation of the only maximal element in $S(s_t)$. Thus if $-s_t+n_is_i \in S$ for $i = 1,2,\ldots,t-1$ then $n_i \leq a_i$ for $i = 1,2,\ldots,t-1$ and $-s_t + \sum_{i=1}^{t-1} n_is_i \notin S$ which shows that S is s_t-supersymmetric.

DEFINITION. A semigroup $S = \langle s_1,s_2,\ldots,s_t \rangle$ is called supersymmetric if it is s_i-supersymmetric for all $i = 1,2,\ldots,t$.

Lemma 15 shows that $\langle s_1,\ldots.s_t \rangle$ is supersymmetric if and only if S is symmetric and $L(s_t)$ represents the elements in $S(s_t)$ for all permutation of the generators $s_1,s_2,\ldots,s_t$. All semigroups $\langle s_1,s_2 \rangle$ are supersymmetric, this follows directly from the given characterization.

LEMMA 16. Let S be a semigroup and s_1,s_2,s_3 three elements in S . Suppose that for some natural numbers a_i,b_i,c_i, $i = 1,2$ we have $-s_1 + b_1s_2 + c_1s_3 \notin S$, $-s_2 + a_2s_1 + c_2s_3 \notin S$, $-s_1 + (b_1+1)s_2 \in S$, $-s_1 + (c_1+1)s_3 \in S$, $-s_2 + (a_2+1)s_1 \in S$ and $-s_2 + (c_2+1)s_3 \in S$. Then $c_1 = c_2$.

PROOF. We have $-s_1 + b_1s_1 + c_1s_3 = -s_1 + (b_1+1)s_2 - s_2 + c_1s_3 \notin S$. If $c_1 > c_2$ we get $-s_2 + c_1s_3 \in S$ hence $-s_1 + (b_1+1)s_2 - s_2 + c_1s_3 \in S$ since $-s_1 + (b_1+1)s_2 \in S$ which gives a contradiction. Thus $c_1 \leq c_2$. The same reasoning on $-s_2 + a_2s_1 + c_2s_3 = -s_2 + (a_2+1)s_1 - s_1 + c_2s_3$ gives $c_2 \leq c_1$, hence the result.

We shall now show that the supersymmetric semigroups can be characterized by their minimal set of generators and then calculate $g(S)$ in terms of these generators.

PROPOSITION 17. Let $S = \langle s_1, s_2, \ldots, s_t \rangle$ be supersymmetric and let $\sum_{\substack{i=1 \\ i \neq j}}^{t} a_{ij} s_i - s_j$ for $j = 1,2,\ldots,t$ be the unique expressions for $g(S)$. Then $a_{i1} = a_{i2} = \ldots = a_{it}$ for $i = 1,2,\ldots,t$, that is we have (with $a_{ij} = b_i$)

$$\begin{pmatrix} -1 & b_2 & b_3 & \ldots & b_t \\ b_1 & -1 & b_3 & \ldots & b_t \\ b_1 & b_2 & -1 & \ldots & b_t \\ \cdot & & & & \\ \cdot & & & & \\ \cdot & & & & \\ b_1 & b_2 & b_3 & \ldots & -1 \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \\ \cdot \\ \cdot \\ \cdot \\ s_t \end{pmatrix} = g(S) \begin{pmatrix} 1 \\ 1 \\ 1 \\ \cdot \\ \cdot \\ \cdot \\ 1 \end{pmatrix}$$

PROOF. That $a_{i1} = a_{i2} = \ldots = a_{it}$ follows directly from Lemma 16.

PROPOSITION 18. The semigroup S minimally generated by $s_1, s_2, \ldots, s_t$ is supersymmetric if and only if there are t pairwise relatively prime numbers $p_1, p_2, \ldots, p_t$ such that $s_i = (\prod_{j=1}^{t} p_j)/p_i$ for $i = 1,2,\ldots,t$.

PROOF. It follows from the formula $g(S) = b_1 s_1 + b_2 s_2 + \ldots + b_{t-1} s_{t-1} - s_t$ and from the unique representation in $L(s_t)$ of the elements in $S(s_t)$ that there are $(b_1+1)(b_2+1)\ldots(b_{t-1}+1)$ different elements in $S(s_t)$. But there are s_t elements in $S(s_t)$. In the same way we see that $s_i = \prod_{j=1}^{t} (b_j+1)/(b_i+1)$ for $i = 1,2,\ldots,t$. Let $p_i = b_i + 1$ for $i = 1,2,\ldots,t$. That the p_i's are relatively prime follows from our standard assumption that the generators have GCD equal to one.

PROPOSITION 19. If $S = \langle (\prod_{i=1}^{t} p_i)/p_1, (\prod_{i=1}^{t} p_i)/p_2, \ldots, (\prod_{i=1}^{t} p_i)/p_t \rangle$ then $g(S) = (t-1) \prod_{i=1}^{t} p_i - \sum_{i=1}^{t} ((\prod_{j=1}^{t} p_j)/p_i)$.

PROOF. We have $g(S) = (p_1-1)(\prod_{i=1}^{t} p_i)/p_1 +$
$+ (p_2-1)(\prod_{i=1}^{t} p_i)/p_2 + \ldots + (p_i-1)(\prod_{i=1}^{t} p_i)/p_{t-1} -$
$(\prod_{i=1}^{t} p_i)/p_t = (t-1)\prod_{i=1}^{t} p_i - \sum_{i=1}^{t}(\prod_{j=1}^{t} p_j)/p_i$.

We shall now study how the elements in a semigroup S are distributed in the interval $[0,g(s)]$.

THEOREM 20. Let S be a semigroup of type t and let $n(S)$ be the number of elements in S which are less than $g(S)$. Then we have

$$(t+1)n(S) \geq g(S) + 1 .$$

PROOF. Let the elements in $S \cap [0,g]$ be $0 = t_1 < t_2 < \ldots < t_N$ (so $N=n(S)$) and let $T_i = \{n \in \mathbb{Z}$, $n \notin S$; $n+s \in S$ for all $s \in S$, $s > t_i\}$. Then we have $T_1 = S'$ and $T_N = \mathbb{N} - S$. We claim that there is an embedding $(T_i - T_{i-1}) \xrightarrow{+t_i} T_1$. Let $n \in T_i - T_{i-1}$, i.e. $n + s \in S$ for all $s \in S$, $s > t_i$ and $n + t_i \notin S$. Then $n + t_i \in T_1$ since $(n+t_i) + s \in S$ as soon as $s > 0$, $s \in S$. This means that the number of elements in $T_i - T_{i-1}$ is less than or equal to the number of elements in T_1 , which gives that the number of elements in T_i is at most i times the number of elements in T_1 $(i \geq 2)$. In particular we have that the number of elements in $T_N = \mathbb{N} - S$ is at most N times the number of elements in $T_1 = S'$, i.e. that $g + 1 - n(S) \leq n(S) \cdot t$.

REMARK. We can compare the above result with a question of Wilf, see [Wi]: Is it true that for a semigroup S generated by k elements that $k \cdot n(S) \geq g(S)+1$ with equality if and only if $S = \langle k,k+1,\ldots,2k-1 \rangle$?

First of all our Lemma 1 shows that there is always equality in Wilf's question if $k=2$. If $k=3$ we have shown that the type is at most 2 (Theorem 11) so Wilf's question can be answered positively in this case. On the

other hand, the note after Theorem 11 shows that our theorem does not imply a positive answer to Wilf's question if $k \geq 4$. Finally we observe that we have equality in Wilf's question for all semigroups $S = \langle k, mk+1, mk+2, mk+3, \ldots, (m+1)k - 1 \rangle$ since then $g(S) = mk - 1$, $n(S) = m$ and the number of generators is k.

ADDENDUM. The result in Proposition 5 has now been improved by J. Backelin who has determined the number of semigroups and symmetric semigroups with $g(S)$ fixed, c.f. "On the number of semigroups of natural numbers", Report no 2, 1986, Math. Dept., University of Stockholm.

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